Ji-hong Min

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KAIST, Daejeon, Republic of Korea

github.com/jihongmin63

Curriculum Vitæ — October 21, 2026

Research Interest

I am interested in understanding the structure and behavior of complex computational systems—from large-scale software under formal specification to machine learning models—through mathematical modeling, property-based testing, and mechanized reasoning.

Education

KAIST (Korea Advanced Institute of Science and Technology) Feb. 2024 – Present

Bachelor of Science in *Mathematical Science* and *Computer Science*

Individually Designed Minor in *Computational Neuroscience*

- Cumulative Grade Average: 97.3% (4.03 / 4.3)

Korea Science Academy of KAIST Feb. 2021 – Feb. 2024

Gifted Science High School

Publications & Presentations

1. Sukyoung Ryu, Seokhun Jeong, Jaehyun Lee, and **Jihong Min**, "Mechanized Specifications for Real-World Programming Languages," *The 47th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, Tutorial, June 2026.

Research Experience

Programming Language Research Group @ KAIST Dec. 2025 – Present

Undergraduate Researcher Advisor: Prof. *Sukyoung Ryu*

- Surveying tools that enable program execution based on formal language specifications. (Link)
- Designing specification-guided PBT for programming language implementations through AI-based synthesis of properties, generators, and Lean proofs. (Link)

Computational Cognitive Neuroscience Lab Jun. 2025 – Dec. 2025

Undergraduate Researcher Advisor: Prof. *Yul HR Kang*

- Proposed a prospect-theory-based RL model for loss-averse exploration behavior. (Link)

Honors and Awards

National Scholarship for Science and Engineering 2024 – Present
Korean government scholarship for outstanding undergraduates in science and engineering.

KAIST Presidential Fellowship 2024 – Present
KAIST fellowship awarded to selected top students.

Skills

Programming Languages	Python, C, OCaml, Rocq, SpecTec
Languages	Korean (Native), English (TOEFL iBT 98, CEFR C1)

Service & Leadership

Republic of Korea Air Force May. 2027 – Feb. 2029 (Expected)
Cyber Systems Operations
- Scheduled for 21 months of mandatory military service in the Electronic Computing specialty.

CS101: Introduction to Programming (KAIST) Fall 2026
Teaching Assistant
- Led weekly lab sessions for ∼80 students; graded assignments and exams.

KAIST Times 2024 – 2025
Reporter, Section Chief
- Covered campus and external affairs for the student newspaper of KAIST and Daejeon Innopolis.